# Worship Ministry Chord Chart App: Product Specification

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## 1. Overview

This app helps a church Worship Ministry team create, edit, store, and share chord charts for songs. Today the team enters chords as plain text, positioning each chord above the lyrics by hand with the spacebar. This is slow and breaks easily when lyrics or fonts change. The app replaces that with a click-to-place chord editor, backed by an AI-assisted draft of lyrics and chords from audio, and clean export to PDF, Word, and plain text.

**Platform:** the app is web-based: the team opens it in a browser, with nothing to install.

## 2. Goals and Non-Goals

### Goals

- Make it fast to build an accurate chord chart without aligning anything by hand.
- Give the team a first draft of lyrics and chords from an uploaded audio file, which they then correct.
- Produce print-ready and shareable charts in PDF, Word, and plain-text formats.
- Keep every song in one organized library the whole team can use.

### Non-Goals (for the first version)

- Public sharing or a public song marketplace.
- Live projection of lyrics to a congregation screen.
- Audio playback, multitrack, or click-track features.
- Perfect, fully automatic chord recognition. The AI output is a draft, not a final answer.

## 3. Users and Roles

- **Worship leader / admin:** creates and edits charts, builds setlists, manages the team and the song library.

- **Musician (band member):** views charts, transposes to a needed key, exports or prints.
- **Contributor (optional):** can add or edit charts but not manage the team.

Open question: does the first version need multiple roles, or can everyone on the team have the same access?

## 4. Scope and Priorities

Priorities use MoSCoW labels: **Must**, **Should**, **Could**, **Later**.

- **Must:** chord chart editor, song library, transposition, PDF export, Word export, plain-text export, import from existing plain-text format.
- **Should:** undo/redo, section management, key and capo settings, team accounts.
- **Could:** AI transcription from mp3 upload, setlists, Nashville number view.
- **Later:** YouTube link transcription, real-time collaboration, offline mode, projection view.

Rationale: the editor and export are useful on their own and carry the least technical risk, so they come first. AI transcription is valuable but uncertain in accuracy, so it follows as a draft generator.

## 5. Functional Requirements

### 5.1 Song Header

Each song has these fields:

- Title (required)
- Writers
- Performed by
- Key (required, used for transposition)
- Optional: capo, tempo (BPM), time signature, CCLI song number, notes

### 5.2 Chord Chart Editor (Must)

This is the core feature. The editor displays lyrics with an empty clickable slot above each word.

- Clicking a slot opens a chord picker anchored to that spot.

- The picker shows the most common chords for the song's key first and allows typing any chord manually.
- A chord can be moved, replaced, or removed with one action.
- Chords can be split to a syllable level when a chord change falls inside a word (for example, the second syllable of a word). Word-level slots are the default.
- Sections are labeled (Intro, Verse 1, Chorus, Bridge, Outro, and custom names) and can be added, renamed, duplicated, and reordered.
- Instrumental or chord-only lines (for example, an intro written as bars with slashes) are supported as a separate line type with no lyrics. Free-text note lines (for example, "Capo 2 for guitar players" or "Repeat twice") are another line type, which can be added anywhere and appear in every export.
- Undo and redo work for every edit.
- Autosave protects against lost work.

### 5.3 Lyrics Entry (Must)

- The user can paste plain lyrics, and the app splits them into sections, lines, and words automatically.
- Lyrics can be edited at any time: words can be retyped, added, or deleted, and lines can be merged or split, without losing chords already placed.

### 5.4 Import of Existing Plain-Text Charts (Must)

The team already has songs in a plain-text format with chords on a line above lyrics and section tags such as [Verse 1] or [Refrain]. The app must convert these into the structured format.

- Section tags in square brackets become sections.
- A line made mostly of chord symbols is treated as a chord line, and the line below is its lyrics.
- Each chord is attached to the word at the matching horizontal position.
- Lines the importer is unsure about are flagged for review, not silently guessed.

### 5.5 Transposition and Key Handling (Must)

- Transposing changes every chord and the song key in one action.
- The app handles sharps and flats correctly and lets the user choose their preference (for example, C# versus Db).

- Capo settings show both the sounding key and the shapes to play.
- Chord parsing supports roots, accidentals, qualities (m, maj7, m7, sus, add, dim, aug), and slash chords.

### 5.6 Song Library (Must)

- A list of all saved songs with search by title, writer, or key.
- Create, duplicate, rename, and delete songs.
- Songs are private to the team's account.

### 5.7 Export (Must for PDF, Word, and plain text)

- PDF export keeps chords aligned above the correct words and is ready to print.
- Word export produces an editable document. Alignment can depend on fonts, so the export should use a fixed layout approach to keep chords positioned correctly.
- Export options for PDF and Word: page size, font size, one or two columns, show or hide section labels, chosen key.
- Optional: export a full setlist as a single file.

**Plain-text export (Must):** one click copies or downloads the chart as plain text, so it can be pasted into another app that only accepts text.

- Section tags in square brackets, such as [Intro], [Verse 1], and [Refrain].
- Each chord line sits directly above its lyric line, lined up with the right word using spaces.
- Chord-only lines, such as an intro written with slashes and bar lines, are kept as written.
- The text is exported in the chosen key, so a transposed chart pastes correctly.
- The format matches the team's current plain-text charts, so the app can also read its own output back in.

### 5.8 AI-Assisted Transcription (Could)

The user uploads an audio file (mp3) and receives a draft chart.

- Lyrics are transcribed from the audio, ideally after separating vocals from instruments.
- Chords are estimated from the audio and aligned to the lyrics.
- The result opens in the editor as a draft, clearly labeled as AI-generated, so the team can correct it.

- The user can edit any word or chord, and corrections do not require re-running the AI.
- Processing status is shown (queued, processing, ready, failed).

Quality expectations: lyrics will need review, and chord accuracy will vary with the recording, especially for complex voicings and live recordings. Success is measured by time saved compared with building a chart from scratch, not by perfect accuracy.

### 5.9 YouTube Link Transcription (Later)

This is deferred. Downloading audio from YouTube conflicts with YouTube's terms of service and may break without notice. It should only be revisited after a review of its terms and a compliant approach.

### 5.10 Setlists (Could)

- Create a setlist for a service date with songs in order.
- Set a key per song for that service.
- Export or print the whole setlist in one action.

## 6. Data Model (High Level)

A song is stored as structured data, not as text with spaces.

- **Song:** title, writers, performer, key, capo, tempo, notes, created and updated times.
- **Section:** name, order, and a list of lines.
- **Line:** type (lyrics, chord-only, or note), with a list of words or the note text.
- **Word:** text, optional chord, optional syllable splits with their own chords.
- **Setlist:** date, name, ordered list of songs with a per-song key.
- **Team and User:** membership and role.

Because chords are tied to words, edits to lyrics never break alignment, and transposition only changes chord names.

## 7. Non-Functional Requirements

- **Usability:** works well on phones and tablets as well as desktop, since musicians often read from tablets.
- **Speed:** opening a song and placing a chord should feel immediate.
- **Reliability:** autosave and version history so a mistake can be undone.

- **Accessibility:** readable fonts, good contrast, adjustable text size.
- **Privacy and security:** only team members can see the team's songs. Uploaded audio is stored securely and can be deleted.
- **Portability:** the team can export all of its data so it is never locked in.

## 8. Copyright and Licensing

- Lyrics and chord charts for most worship songs are copyrighted. Churches commonly rely on a license such as CCLI, so the team should confirm what its own license allows.
- Songs are kept private to the team's account, not published publicly.
- The app can store a CCLI song number per song to help with reporting.
- This section is a planning note, not legal advice. The church should check its license terms directly.

## 9. Milestones (Editor First)

The chord chart editor is built first, in five phases. Each phase ends with something the team can try, and later features wait until the editor meets the definition of done below.

1. **Foundation:** the song data model (song, section, line, word, chord), the chord parser, and transposition logic.
   - Done when: parsing and transposition give correct results on a set of sample songs, including sharps, flats, and slash chords.
2. **Basic editor:** paste lyrics and have them split into sections, lines, and words; click above a word to place a chord; move, replace, or remove chords; label sections.
   - Done when: a full song can be charted by clicking alone, with no spacebar alignment.
3. **Safe editing:** undo and redo, autosave, chord-only lines, optional syllable splits, and lyric edits that keep existing chords.
   - Done when: closing and reopening a song shows it exactly as left, and any mistake can be undone.
4. **Library and plain text:** save, open, search, and duplicate songs; import the team's plain-text format with unsure lines flagged; export plain text in the chosen key.
   - Done when: importing a plain-text song and exporting it again puts every chord above the same word as before.

5. **Team trial:** real team members chart real songs on a laptop and a tablet, and the issues they find are fixed.
    - Done when: the team agrees that charting a song is faster than the spacebar method.

### Definition of done for the editor

The editor counts as done when all of these are true:

- A new song can be created from pasted lyrics and fully charted by clicking, with no spacebar alignment.
- Chords can be placed, moved, replaced, and removed, and each action can be undone and redone.
- Transposing changes every chord and the song key correctly, with a choice of sharps or flats.
- Section labels and chord-only lines (such as an intro with bar lines) work.
- Songs autosave and reopen exactly as left.
- Existing plain-text songs import with unsure lines flagged, and plain-text export puts every chord back above the same word.
- It works in a laptop browser and on a tablet.
- The team has charted at least five real songs from its own repertoire with it.

### After the editor

These follow only once the editor is done:

1. **Export:** PDF and Word export.
2. **Team features:** accounts, roles, private sharing.
3. **AI draft:** mp3 upload, transcription, draft opens in the editor.
4. **Extras:** setlists and any Later items.

## 10. Risks

- AI chord accuracy may fall short of expectations. Mitigation: treat output as a draft and make correction fast.
- Word and plain-text alignment may differ across devices and apps because of fonts. Mitigation: use a fixed layout for Word, and test all three outputs in the programs the team actually uses.

- Copyright and licensing misunderstandings. Mitigation: keep songs private and confirm the church's license.
- Scope growth. Mitigation: hold to the Must list for the first release.

## 11. Open Questions

- Who will use the app at first: one church team or several churches?
- How many users and songs are expected?
- Should the web app be fully usable on phones and tablets from day one, and does it need to work offline?
- Is a login system needed from day one?
- Who will build it, and what is the target timeline and budget?
- Which languages must the app support for lyrics and interface?
- Is the current plain-text format the only import source, or should other formats (such as ChordPro) be supported?
- Are there features from the team's current tools that must carry over?